



Latest developments in the ALA

Living Atlases Workshop

20 September 2026

TDWG 2026, Oslo Norway



The ALA is the AU node of GBIF, and receives support from the Australian Government through the National Collaborative Research Infrastructure Strategy (NCRIS) and is hosted by the [Commonwealth Scientific and Industrial Research Organisation \(CSIRO\)](#).

Agenda

- Galah
- Quail plugin
- Pipelines fork
- Parquet exports
- UX/UI
- AI in ALA
- Non-core apps
- Q&A



Ghost Fungus (*Omphalotus nidiformis*)
Photo Credit rogerp CC BY NC



Dainty Green Tree Frog (*Litoria gracilentia*)
Photo Credit Nick Lambert CC BY NC

Galah



Galah

- galah is an interface to biodiversity data hosted by the ALA(, GBIF and LA nodes)
- Available for R and Python



Galah

- Upcoming version:
 - First time JWT token-based authentication is implemented outside the ALA
 - Select atlas instance at runtime

```
request_data(from = "Spain") |>  
count() |>  
collect()
```



Quail plugin

Quail plugin

- Quail is a plugin for QGIS
- Quail enables users to locate and download species occurrence records
 - Queries
 - Record type
 - Status
 - Data quality profiles
 - Data read as vector layer



Quail plugin

- Getting Started
- https://quail.ala.org.au/getting_started.html



Quail plugin

Quail

Atlas of Living Australia
ala.org.au

Visit the ALA Open Documentation Contact Your Atlas

Taxonomy Date Spatial Authoritative Lists Other

Enter Names (separated by ';'):

Upload Species List

Check Taxonomic Backbone

email: Reason:

atlas:

☐ Create a DOI DOI:

Check Number of Occurrences

Import Occurrence Records

Australia
Austria
Brazil
Flanders
GBIF
Kew Gardens Data Portal
Spain
Sweden
United Kingdom



Quail plugin

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Taxonomy Date Spatial Authoritative Lists Other

Enter Names (separated by ';'):

Vulpes vulpes

Upload Species List

Check Taxonomic Backbone

email: Reason:

atlas: ☐ Create a DOI

Check Number of Occurrences

Get Species List With Counts and Statuses Import Occurrence Records



Quail plugin

Quail

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Taxonomy Date Spatial Authoritative Lists Other

Start Date End Date

January 2026 August 2026

	Mon	Tue	Wed	Thu	Fri	Sat	Sun
1	29	30	31	1	2	3	4
2	5	6	7	8	9	10	11
3	12	13	14	15	16	17	18
4	19	20	21	22	23	24	25
5	26	27	28	29	30	31	1
6	2	3	4	5	6	7	8

	Mon	Tue	Wed	Thu	Fri	Sat	Sun
31	27	28	29	30	31	1	2
32	3	4	5	6	7	8	9
33	10	11	12	13	14	15	16
34	17	18	19	20	21	22	23
35	24	25	26	27	28	29	30
36	31	1	2	3	4	5	6

email: Reason:

atlas: ☐ Create a DOI

Check Number of Occurrences

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Quail plugin

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2	5	6	7	8	9		5	6	7	8	9
3	12	13	14	15	16		12	13	14	15	16
4	19	20	21	22	23		19	20	21	22	23
5	26	27	28	29	30		26	27	28	29	30
6	2	3	4	5	6		2	3	4	5	6

 Your query will return 32012 records.

Cancel OK

email:  ala4r@ala.org.au Reason:  -- None --

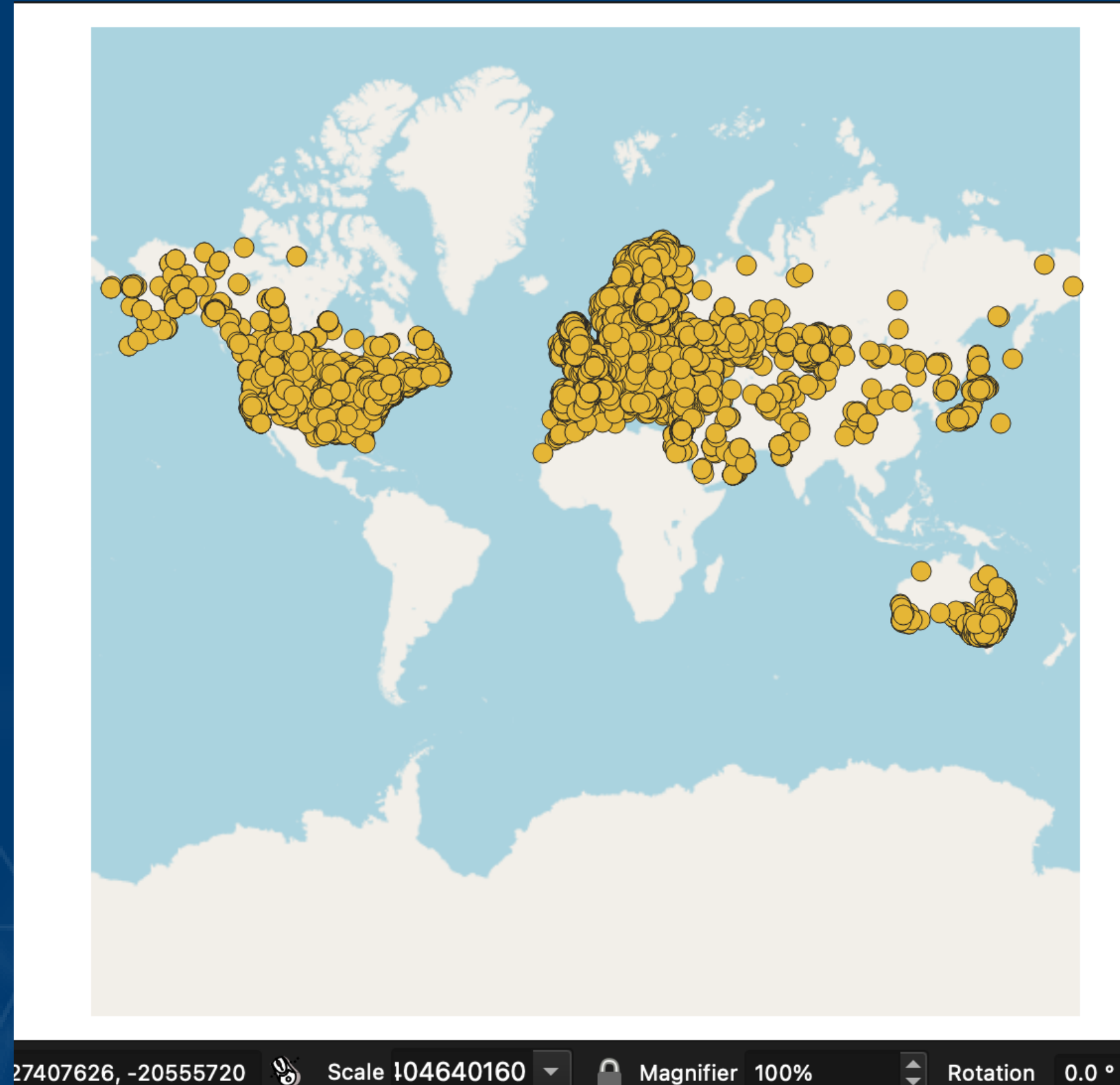
atlas: GBIF ☐ Create a DOI  DOI:

Check Number of Occurrences

Get Species List With Counts and Statuses Import Occurrence Records



Quail plugin

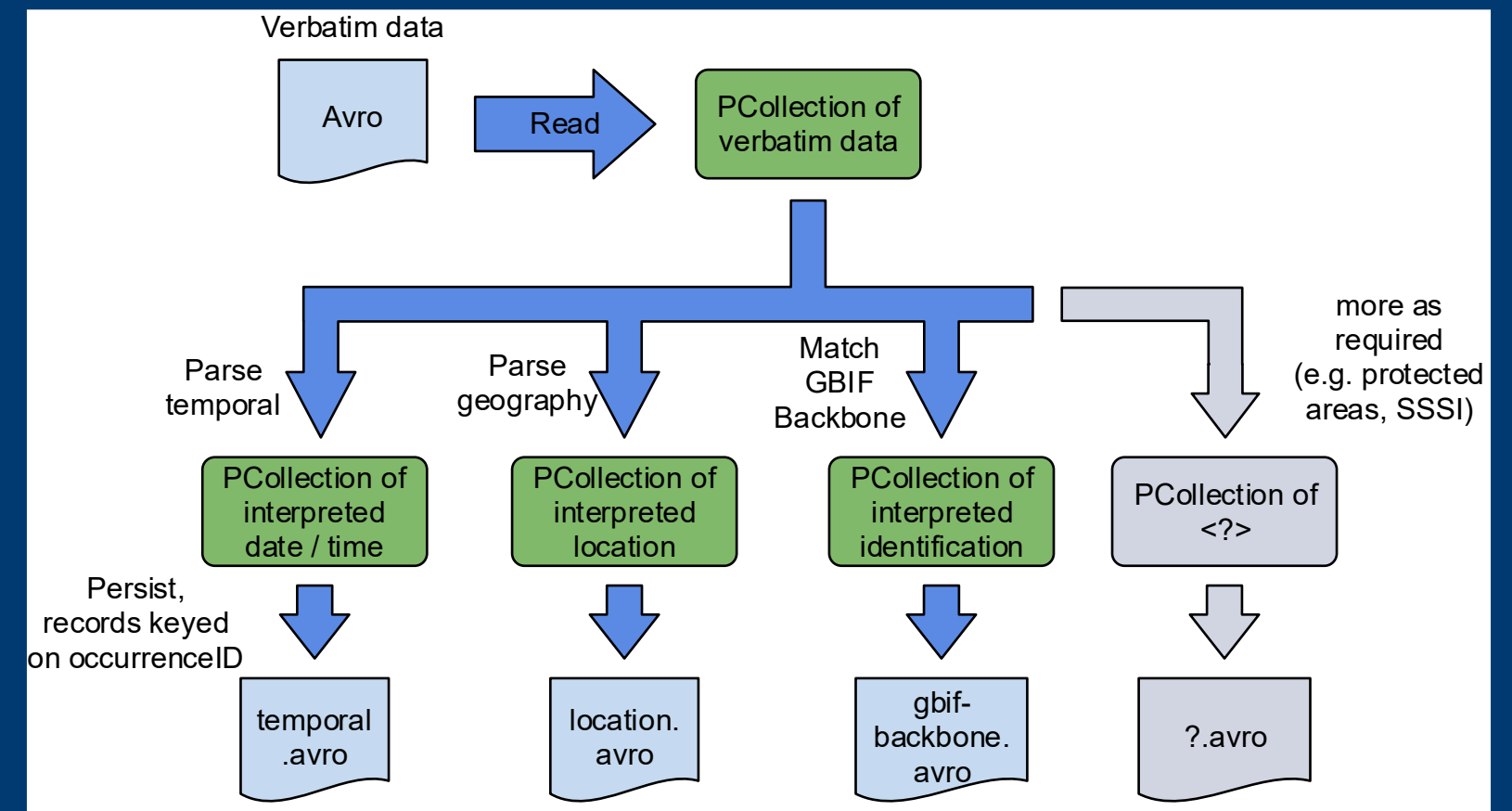


Pipelines work



Pipelines fork

- Forked the GBIF repository into ALA GitHub
- Updates to pipelines pipelines-airflow for EMR 7.x (running on Spark 3.5.6 and Java 17)
- Adding tests for avros using airflow and pyspark to identify differences with current prod.
- Include latest GBIF vocabularies including Event Types

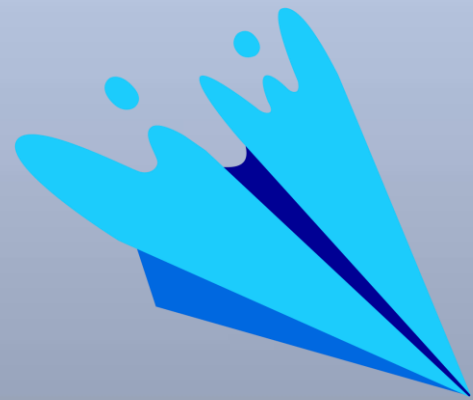


Parquet exports



Parquet exports

Why?



There is currently no practical way for internal or external users to download very large numbers of occurrence records through the existing API-based approach.

It is capped at 50 million records.

Parquet exports

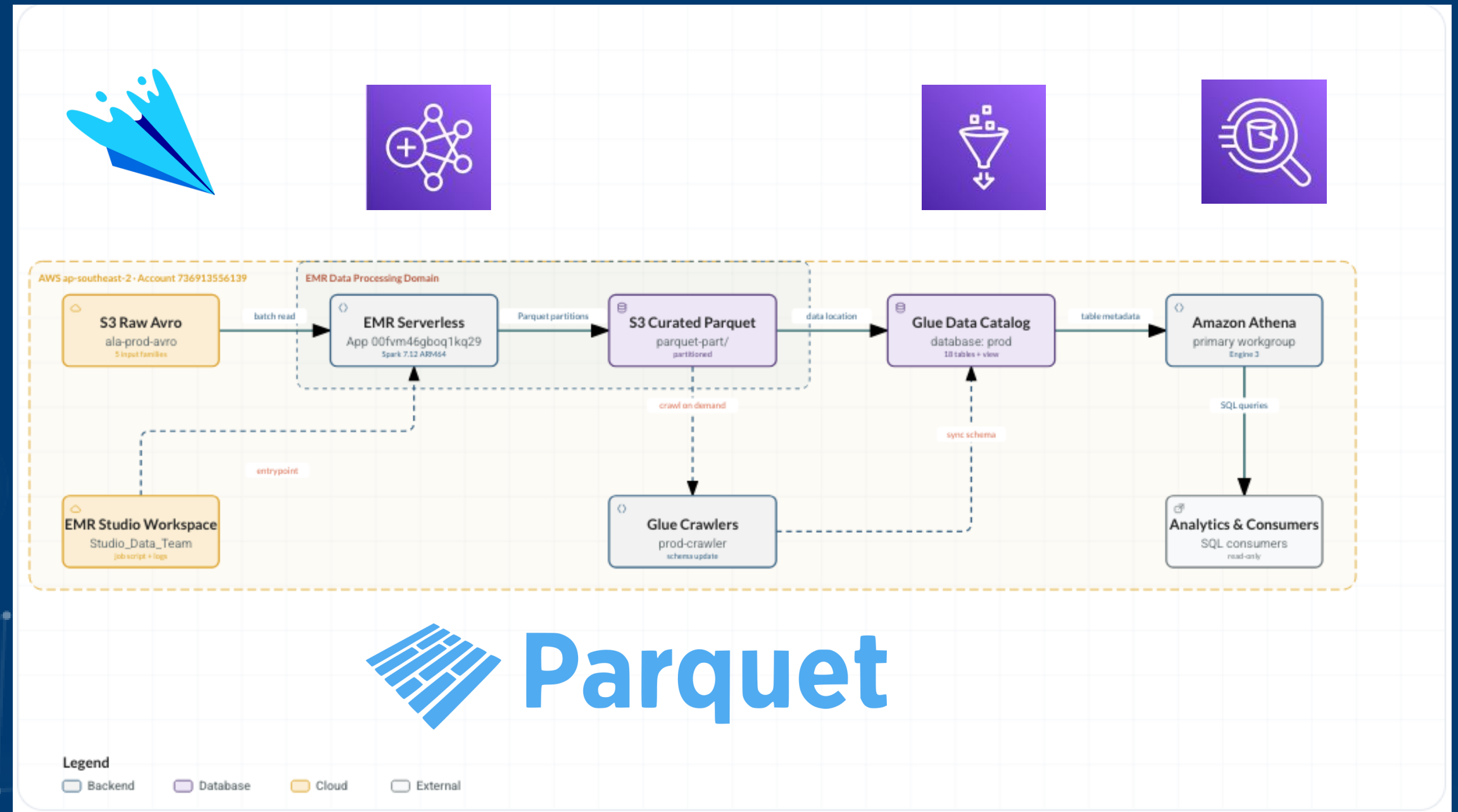
What?

- Produce regular, comprehensive Parquet snapshots of occurrence data.
- Improve accessibility of large biodiversity datasets for ALA users.
- Ensure export reliability, integrity, and repeatability.
- Support scalable download workflows.
- Improve documentation and guidance for end users.
- Align export outputs with downstream query and download use cases.
- ALA internal data interrogation across all occurrences and taxonomic data.

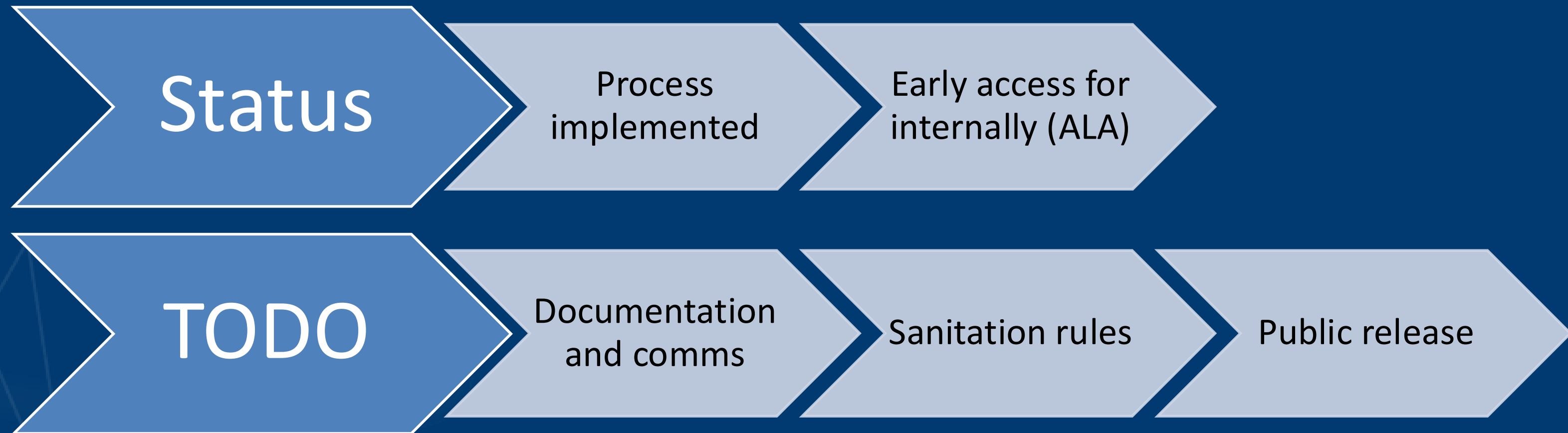
Parquet exports

How?

- Interpreted AVRO
- EMR
- Glue
- Parquet
- Athena



Parquet exports



Parquet exports

- What are the taxonomy values used and match type across 180 million occurrence records

Purpose: Validate name matching service across 2+ million different queries

The screenshot displays a data query interface with a sidebar on the left and a main query editor on the right. The sidebar includes sections for 'Data' (Data source: AwsDataCatalog, Catalog: None, Database: prod), 'Tables and views' (with a 'Create' button and a search bar), and a list of 18 tables (additional, cl, el, event, geologicalcontext, identification, location, materialsample, measurementorfact, occ, occurrence, organism, outlier, record) all marked as 'Partitioned'. The main query editor shows two queries; Query 1 is selected and contains a SQL query that selects distinct taxonomy and match type information from a 'taxon' table, joined with an 'additional' table. Below the query editor, there are buttons for 'Run again', 'Explain', 'Cancel', 'Clear', and 'Create'. The 'Query results' tab is active, showing a 'Completed' status with a progress bar. It also displays performance metrics: 'Time in queue: 95 ms', 'Run time: 1 min 22.771 sec', and 'Data scanned: 13.93 GB'. Below these metrics, there are buttons for 'Copy' and 'Download results CSV'. The results table shows columns for various taxonomy levels (rawScientificName, rawKingdom, rawPhylum, rawClass, rawOrder) and lists the first three rows of data.

Editor Recent queries Saved queries Query settings Workgroup primary

Data

Data source: AwsDataCatalog

Catalog: None

Database: prod

Tables and views Create Filter tables and views

Tables (18)

- additional Partitioned
- cl Partitioned
- el Partitioned
- event Partitioned
- geologicalcontext Partitioned
- identification Partitioned
- location Partitioned
- materialsample Partitioned
- measurementorfact Partitioned
- occ Partitioned
- occurrence Partitioned
- organism Partitioned
- outlier Partitioned
- record Partitioned

Query 1 : X Query 2 : X

```
1 Select distinct raw_scientificName as rawScientificName,
2 raw_kingdom as rawKingdom,
3 raw_phylum as rawPhylum,
4 raw_class as rawClass,
5 raw_order as rawOrder,
6 raw_family as rawFamily,
7 raw_genus as rawGenus,
8 raw_specificepithet as rawSpecificEpithet,
9 raw_infraspecificepithet as rawInfraspecificEpithet,
10 verbatimtaxonrank as verbatimtaxonrank,
11 specificepithet,
12 infraspecificepithet,
13 raw_species as rawSpecies,
14 matchtype as matchType,
15 scientificName,
16 taxonconceptid,
17 raw_vernacularname as rawvernacularname, raw_scientificnameauthorship as rawscientificnameauthorship, taxonid as rawtaxonid
   from taxon t, additional a where t.id_prefix=a.id_prefix and t.dataresourceuid = a.dataresourceuid and t.id = a.id
```

SQL Ln 17, Col 239

Run again Explain Cancel Clear Create

Reuse query results up to 60 minutes ago

Query results Query stats

Completed Time in queue: 95 ms Run time: 1 min 22.771 sec Data scanned: 13.93 GB

Results (2,028,047) Copy Download results CSV

Search rows

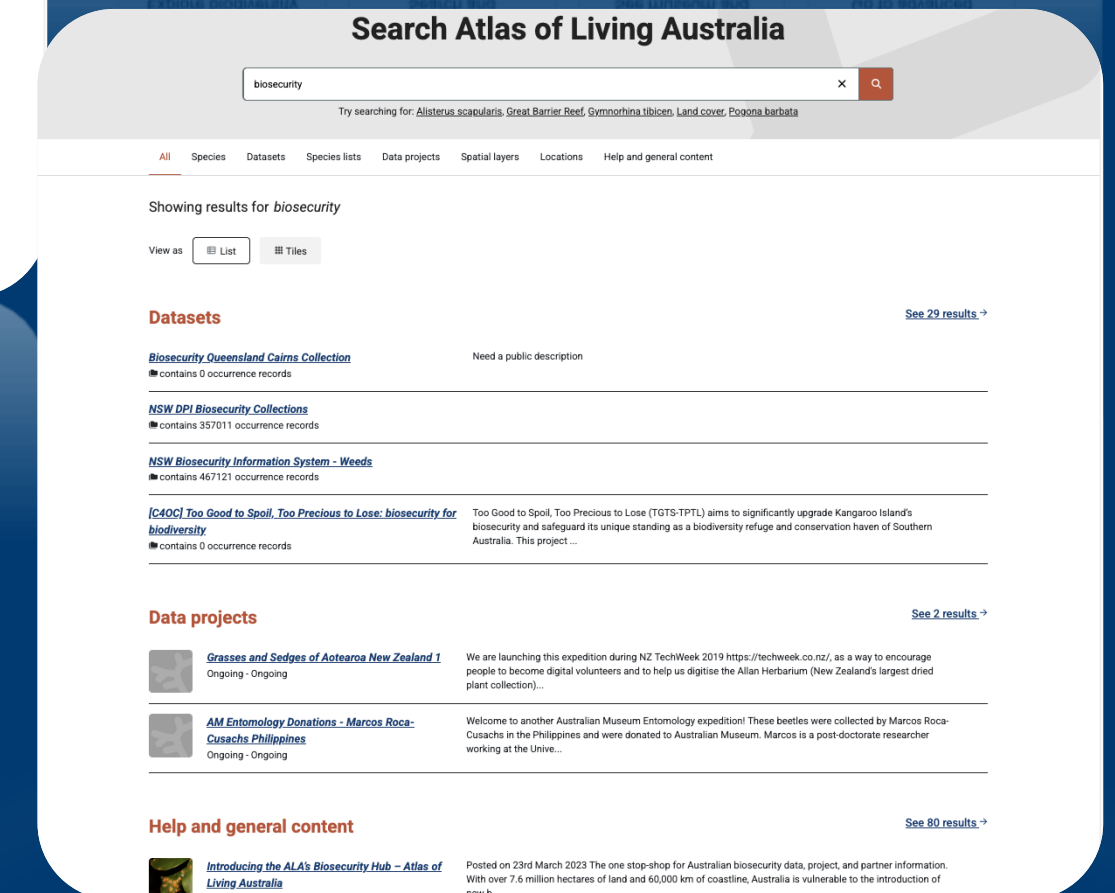
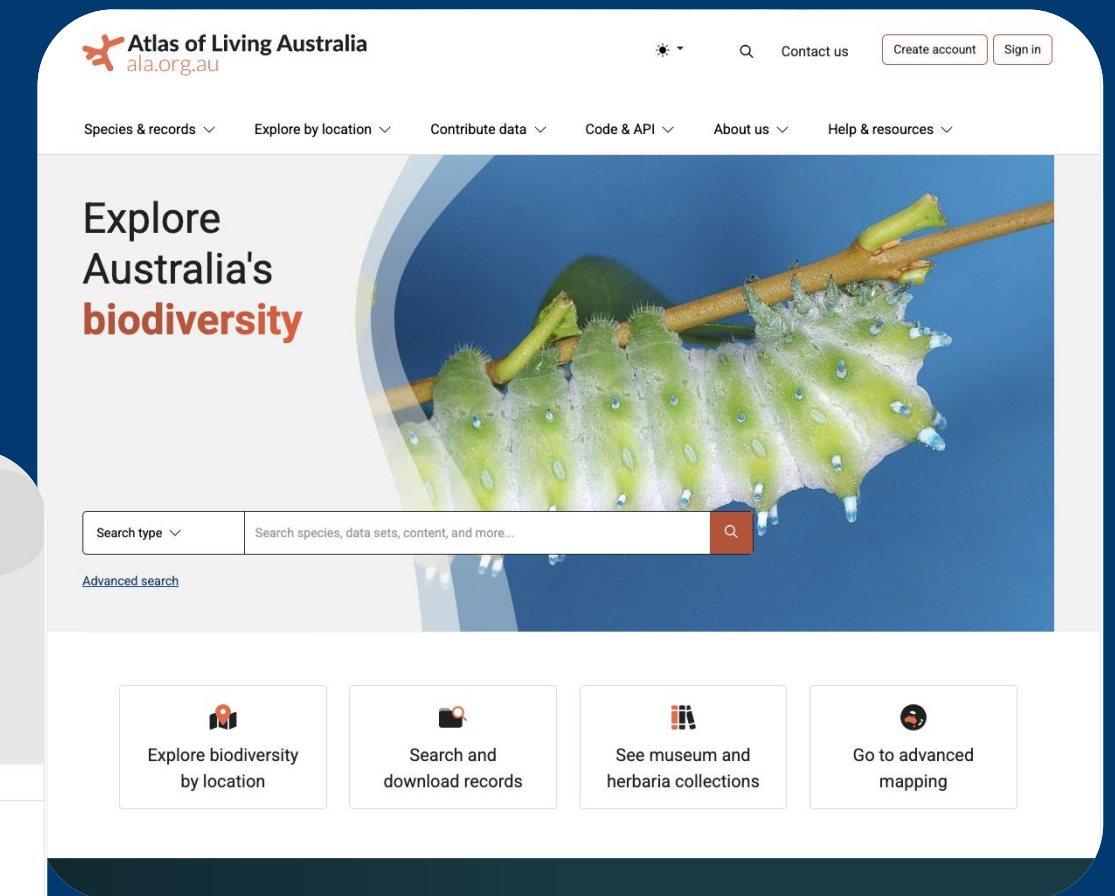
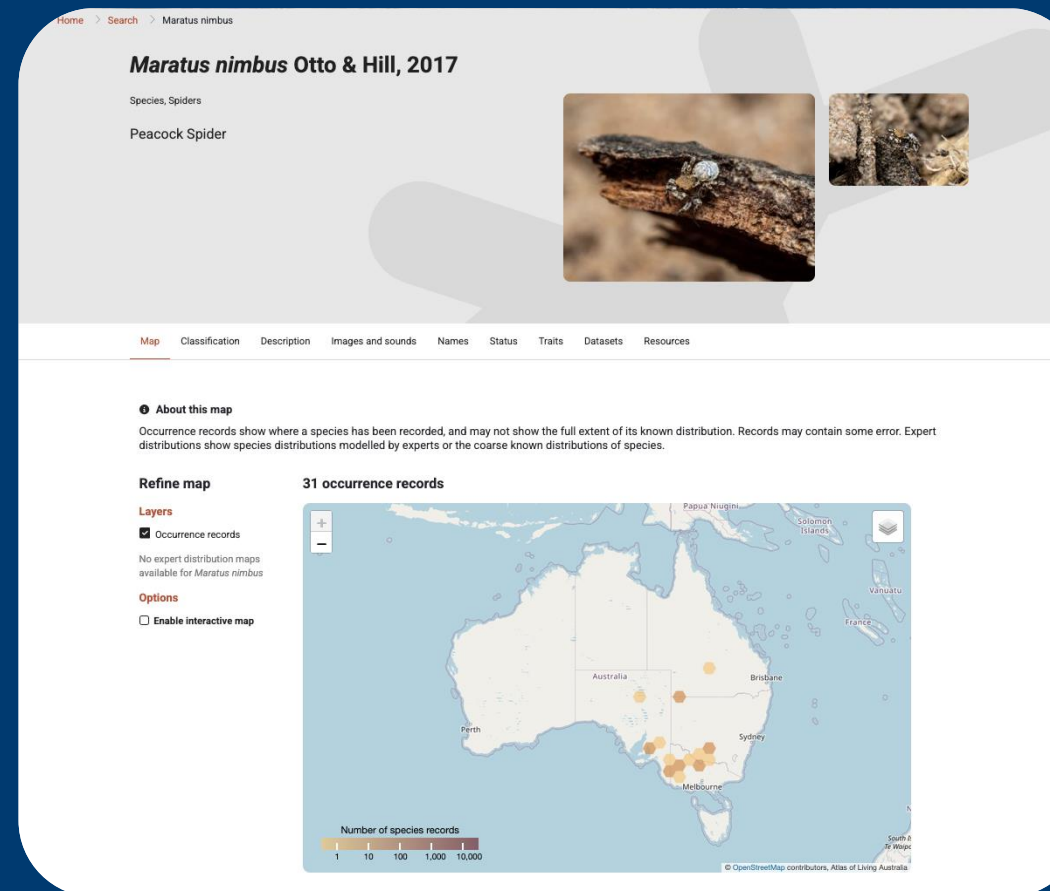
#	rawScientificName	rawKingdom	rawPhylum	rawClass	rawOrder
1	Menegazzia fortuita Elix & P.M.McCarthy				
2	Tayloria octoblepharum (Hook.) Mitt.				
3	Eucalyptus amygdalina				

UX/UI



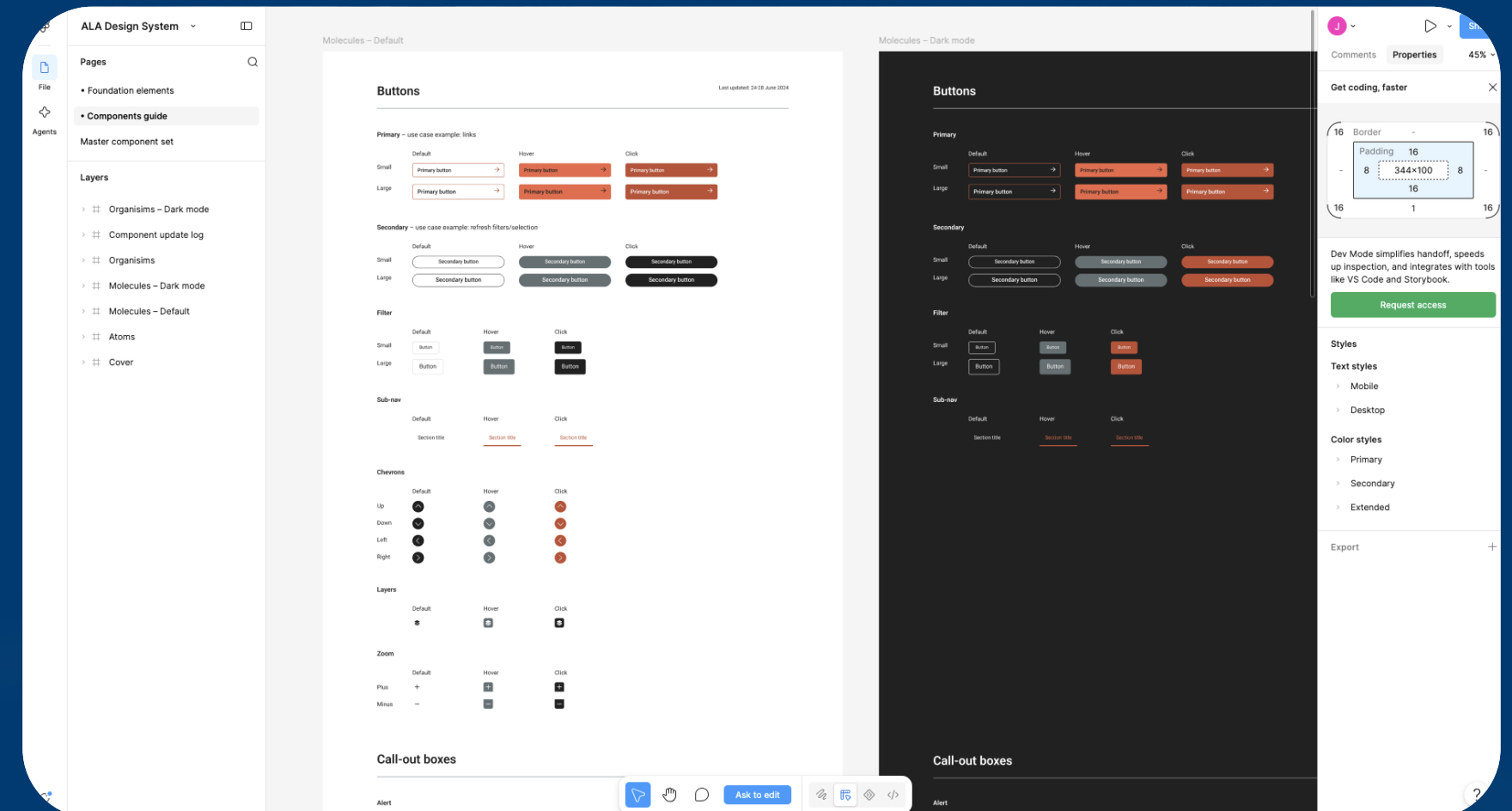
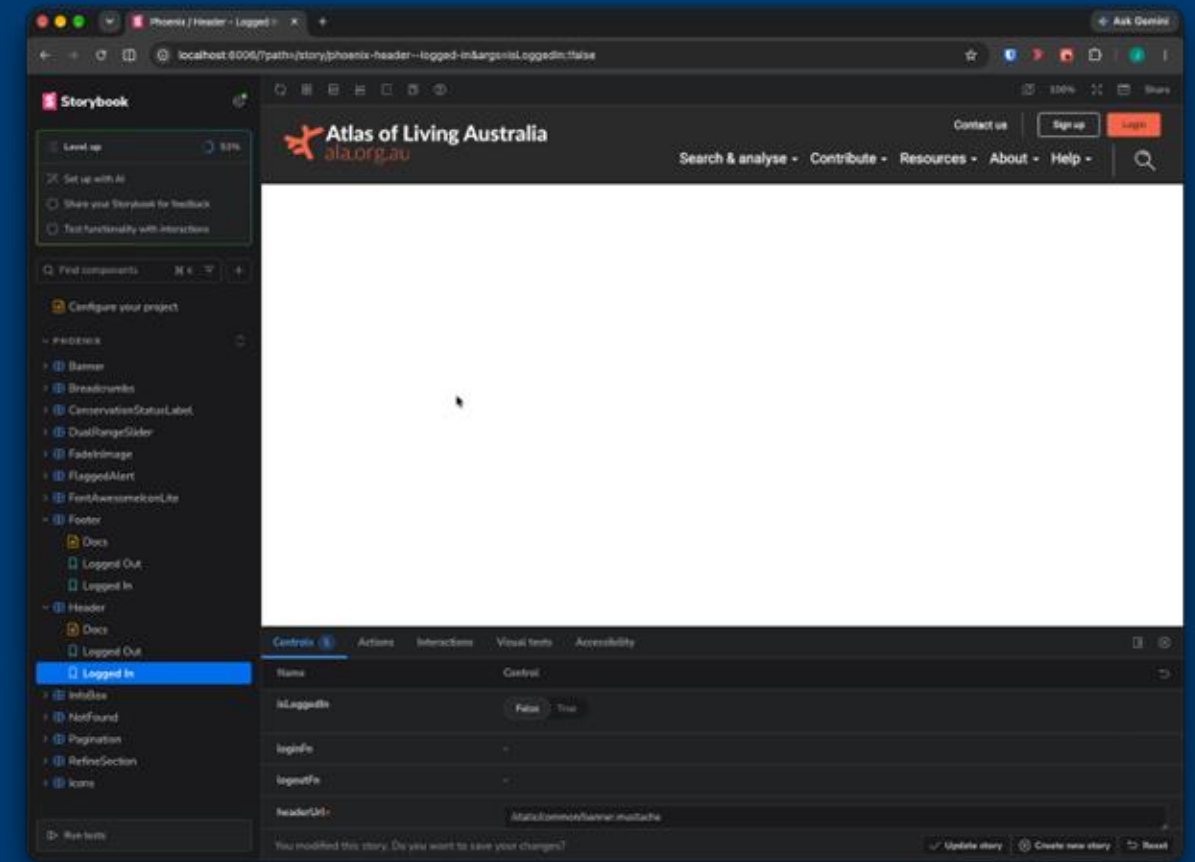
UX/UI

- Establish UX/UI practice in ALA to consistently improve user experience for users of the ALA
- Refresh UI for Species Pages, global search and Homepage



UX/UI

- Operationalise existing designs
- NDRI project funding

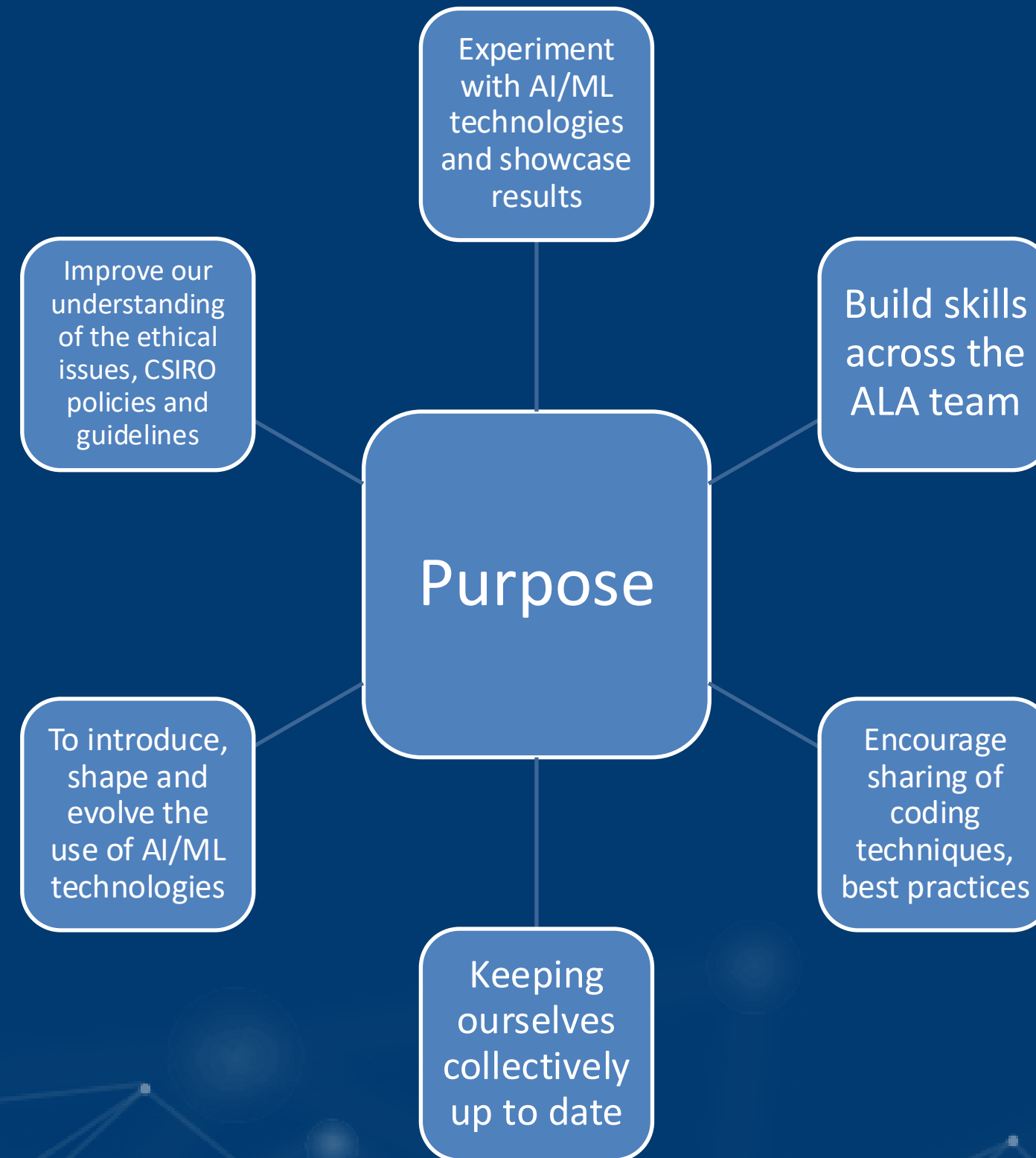


AI in ALA



AI/ML Community of Practice

Established in Feb 2026



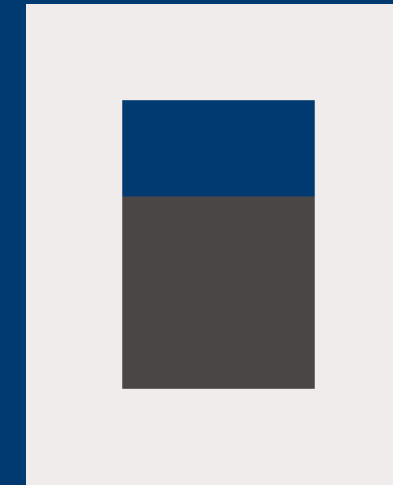
AI/ML Community of Practice

Sessions so far

Aug 26	Ethics, privacy and equity in AI adoption
May 26	1. Biodiversity data and AI-assisted workflows 2. Working with Claude Code
Apr 26	AI use cases: 1. Using skills and context to build your own agent 2. NotebookLM 3. Refactoring a Java/Spring app with OpenCode 4. Bridging Biodiversity Silos: AI Integration of Species Lists, Traits, Images and Spatial Data
Feb 26	Integrating Large Language Model and ALA data

Agentic code reviews

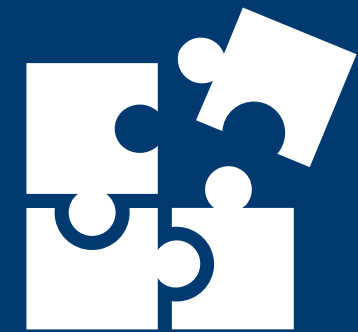
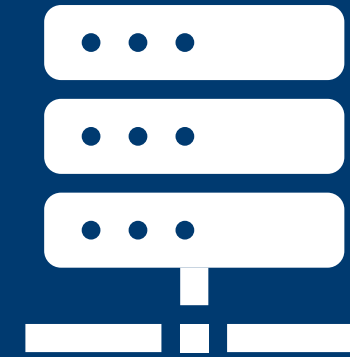
- Why?
Increase of security attacks and breaches from AI agents
- Open source is particularly vulnerable
- Our agentic code review contains tools and resources for the OpenCode AI assistant to perform security and performance audits on ALA code repos



opencode

ALA API MCP Server

- A Model Context Protocol (MCP) server for interacting with the Atlas of Living Australia (ALA) API.



Disclaimer: This project is only intended for demonstration and educational purposes for the internal ALA team.

Non-core apps

Non-core apps

- ARGGA is closing
- BioCollect's PWA improvements

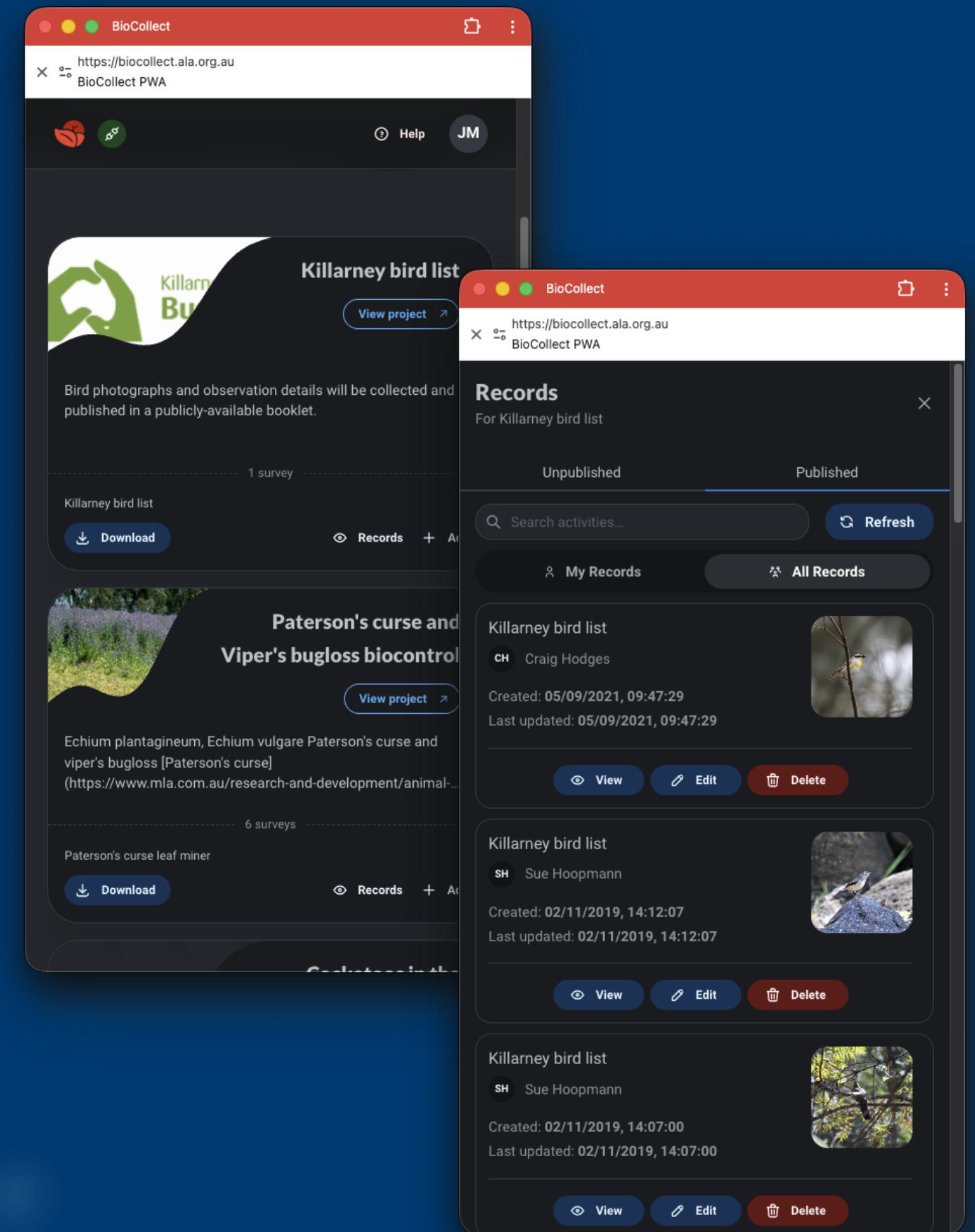
ARGA

- Prototype that improved the discoverability, accessibility and interpretation of genomic data for Australian species.
- Reached a global audience across 117 countries.
- Demonstrated successful integration of genomic and biodiversity data, linking sequences to organisms, specimens and taxonomy.



• Biocollect PWA

- AVSB Website updates
- Profiles tool uplift
- BioCollect's PWA improvements



Q&A





Yellow-footed Antechinus (*Antechinus flavipes*)
Photo Credit Martin Stokes CC BY NC



Atlas of Living Australia
ala.org.au

**Thank you and
the ALA team!**

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